

Question:

Shock causes:

Answer Choices:

- A. Stagnant hypoxia
- B. Anemic hypoxia
- C. Hypoxic hypoxia
- D. Histotoxic hypoxia

Reasoning + Answer:

Shock leads to inadequate blood flow to tissues, resulting in insufficient oxygen delivery to cells. This is primarily due to reduced perfusion, not a decrease in oxygen content of the blood or impaired cellular utilization.

Let's evaluate each option:

- A. Stagnant hypoxia This occurs when blood flow to tissues is reduced (e.g., due to poor perfusion), leading to oxygen deprivation despite adequate oxygen content in the blood. This is exactly what happens in shock, where circulation is impaired and blood flow is stagnant. This is the most appropriate answer.
- B. Anemic hypoxia This results from a decrease in hemoglobin or red blood cells (e.g., hemorrhage, hemolytic anemia), leading to reduced oxygen-carrying capacity. While hemorrhagic shock may involve anemia, the primary mechanism of hypoxia in shock is not anemia per se, but rather poor perfusion.
- C. Hypoxic hypoxia This occurs when oxygen content in the blood is low due to low partial pressure of oxygen (e.g., in pulmonary disorders). This is not the primary mechanism in shock.
- D. Histotoxic hypoxia This occurs when cells cannot utilize oxygen (e.g., cyanide poisoning). It is not related to shock.

In shock, the main problem is that tissues receive insufficient blood flow, leading to stagnant hypoxia. The blood may have normal oxygen content, but it's not reaching the tissues due to poor circulation.

<answer>

A. Stagnant hypoxia

</answer>